



Contents lists available at ScienceDirect

Journal of the American Pharmacists Association

journal homepage: www.japha.org

REVIEW

Prevalence of sodium-glucose transporter 2 inhibitor-associated diabetic ketoacidosis in real-world data: A systematic review and meta-analysis

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ARTICLE INFO

Article history:

Received 12 June 2023

Accepted 9 October 2023

Available online 14 October 2023

ABSTRACT

Background: The U.S. Food and Drug Administration (FDA) revised the labels of sodium-glucose transporter 2 (SGLT2) inhibitors in December 2015 to inform users regarding the risk of diabetic ketoacidosis (DKA). As more drugs of this class are approved and their indications are expanded, this serious adverse effect has been increasingly reported.

Objective: This review evaluated observational studies to inform the prevalence of SGLT2-inhibitor-associated DKA compared with other antihyperglycemic agents.

Methods: A systematic review was conducted in PubMed and EMBASE until 19 July 2022 (PROSPERO: CRD42022385425). We included published retrospective cohort active comparator/new user (ACNU) and prevalent new user studies assessing SGLT2-inhibitor-associated DKA prevalence in adult patients with type 2 diabetes mellitus (T2DM) against active comparators. We excluded studies which lacked 1:1 propensity score matching. The JBI Checklist for Cohort Studies guided the risk-of-bias assessments. Meta-analysis was conducted based on the inverse variance method in R software.

Results: Sixteen studies with a sample of 2,956,100 nonunique patients met the inclusion criteria. Most studies were conducted in North America ($n = 9$) and adopted the ACNU design ($n = 15$). Meta-analysis of 14 studies identified 33% higher DKA risk associated with SGLT2 inhibitors ($HR = 1.33$, 95% CI: 1.14–1.55, $P < 0.01$). Meta-regression analysis identified the study location ($P = 0.02$), analysis principle ($P < 0.001$), exclusion of chronic comorbidities ($P = 0.007$), and canagliflozin ($P = 0.04$) as significant moderator variables.

Conclusions: Despite limitations related to heterogeneity, generalizability, and misclassification, the results of this study show that SGLT2 inhibitors increase the prevalence of DKA among adult T2DM patients in the real world. The findings supplement evidence from randomized controlled trials (RCTs) and call for continued vigilance.

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Background

Diabetes mellitus (DM) may be considered one of the most significant epidemics in human history.¹ Persistent high blood

Funding: This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Registration: The International Prospective Register of Systematic Reviews reference number is CRD42022385425.

Ethics approval statement: This study does not involve human participants or animal subjects.

Data transparency: Data were extracted from published articles, all of which are available and accessible.

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glucose levels in patients with diabetes progressively result in multiple-organ injury, impairment and loss of function, with the impact being more detrimental in the eyes, kidneys, nerves, heart, and vessels.² DM, like other chronic diseases, is often understated in terms of its impact on health and disability, and in terms of its economic burden.³ Over the past decades, researchers and major pharmaceutical companies have been searching for solutions to address the need for novel, more effective antidiabetic medications to achieve desired glycemic targets.

The European Commission and the U.S. Food and Drug Administration (FDA) have approved 4 sodium-glucose transporter 2 (SGLT2) inhibitors to enhance glycemic control in adult T2DM patients. Between November 2012 and August 2014, the European Commission and the FDA each approved dapagliflozin, canagliflozin, and empagliflozin.^{4–9} Ertugliflozin

Key Points**Background:**

- Randomized controlled trials (RCTs) show that sodium-glucose transporter 2 (SGLT2) inhibitors augment diabetic ketoacidosis (DKA) risk in patients with type 2 diabetes mellitus (T2DM).
- The indications of SGLT2 inhibitors continue to expand to include the treatment of heart failure and kidney disease, but it is not clear whether the class is linked to real-world DKA risk.

Findings:

- Meta-analysis of 14 primarily noncommercial, active comparator/new user (ACNU) cohort studies finds SGLT2 inhibitors are associated with a 33% higher risk of DKA.
- The absolute rate of DKA in the real world is between 0.6 and 6.3 per 1000 person-years.
- Studies that are conducted in North America and Scandinavia, employ as-treated or per-protocol analysis principles, utilize chronic-illness exclusion criteria, and observe predominately canagliflozin-treated cohorts tend to report higher DKA risk with SGLT2 inhibitor treatment.
- Pharmacists should stay vigilant and educate patients and colleagues on the risk of this serious but rare adverse effect, as well as the value proposition for using SGLT2 inhibitors.

was approved for use in adult patients with type 2 diabetes mellitus (T2DM) in the USA and the European Union in December 2017 and March 2018, respectively.^{10,11} Dapagliflozin was approved for use in some patients with type 1 diabetes mellitus (T1DM) by the European Commission as an adjunct to insulin in March 2019, but this license was withdrawn in October 2021.¹² Subsequently, SGLT2 inhibitors have been incorporated in the 2020 American Diabetes Association's diabetes care guidelines as second-line treatment for T2DM, regardless of HbA1c levels.¹³

Pharmacists who work at the same location as primary care providers to manage T2DM can positively influence prescription practices, even indirectly, to favor SGLT2 inhibitors and the overall use of novel antihyperglycemic agents when clinically indicated.^{14,15} SGLT2 inhibitors lower blood glucose levels by blocking the SGLT2 cotransporter proteins in the proximal convoluted tubule segment S1 of the kidneys.¹⁶ This process prevents the re-absorption of up to 90% of the filtered D-glucose,¹⁷ and is independent of β -cell function and insulin sensitivity.¹⁸ Interestingly, this mimics the loss of glucose in the urine observed in familial renal glycosuria, a rare hereditary disorder caused by mutated SLC5A2,¹⁹ which is the gene that encodes SGLT2 on chromosome 16p11.2.²⁰ SGLT2 inhibitor use creates persistent mild hyperketonemia,²¹ possibly because of increased lipolysis and ketone reabsorption and a reduced insulin-glucagon ratio.²² In December 2015, the FDA

revised the labels of SGLT2 inhibitors to include warnings about their serious side effects, namely diabetic ketoacidosis (DKA) and serious urinary tract infections (UTIs).²³

Surprisingly, patients who can benefit the most from SGLT2 inhibitors are less likely to use them.²⁴ Pharmacists have an opportunity to correct this paradox by playing a central role in the utilization of SGLT2 inhibitors.²⁵ Educating patients about the uncommon but serious side effects of SGLT2 inhibitors, like DKA, could be lifesaving and cost-effective.²⁶ Pharmacists are in a good position to increase patient awareness about the risks associated with SGLT2 inhibitors. They can offer quality counselling by staying informed about the benefits and potential risks associated with emerging medication classes. The mechanisms by which SGLT2 inhibitors cause DKA are not fully understood. Recent animal experiments suggest that the outcome is strongly mediated by insulinopenia and dehydration.²⁷ Certain patient characteristics and factors may predispose patients who take SGLT2 inhibitors to DKA. These include being middle-aged or female, having a diagnosis of T2DM, using SGLT2 inhibitors off-label in T1DM, undergoing surgery, reducing insulin doses, experiencing an infection or illness, changing the diet or medications, enduring an extended fast, experiencing dehydration, taking a high dose of SGLT2 inhibitors, and specifically taking canagliflozin.^{28–32} Yet, about 22% of the DKA cases involving SGLT2 inhibitors may not be caused by an obvious precipitating factor,²⁹ and up to 89% of the cases are euglycemic.³⁰

Recent systematic reviews of randomized controlled trials (RCTs) concluded that SGLT2 inhibitors augmented DKA risk in patients with T2DM.^{33–36} As RCTs are usually conducted in highly selected populations, epidemiological evidence is needed to determine whether the risk is greater in treated populations at large. A recent systematic review of observational studies attempted to make this determination. However, the authors were unable to provide a conclusive answer regarding SGLT2-inhibitor-associated DKA risk as their review was limited to 5 studies of varying designs and did not include publications after December 2020.³⁷ A subsequent systematic review included studies using the active comparator, new-user (ACNU) design³⁶ but it had several limitations, such as covering a shorter time period and only analyzing data from 5 studies. Additionally, the data analyzed may have included non-peer-reviewed information.^{38,39}

Objective

This study was an updated, more comprehensive systematic review of the data on the association between SGLT2-inhibitor exposure and DKA in patients with T2DM using observational population-based pharmacoepidemiologic studies. Our primary objective was to assess the association between SGLT2 inhibitors and DKA against other antihyperglycemic agents, using high-quality evidence. The secondary objective was to identify potential variables that influenced the risk of DKA in patients treated with this novel class.

Methods*Protocol and registration*

We conducted and reported this meta-analysis following both MOOSE (Meta-analysis of Observational Studies in

Epidemiology) and PRISMA (Preferred Reporting Items for Systematic reviews and Meta-Analyses) guidelines.^{40,41} The eligibility criteria for study inclusion were determined based on the PICO framework as follows:

- Population: Adult patients aged 18 years or above with T2DM
- Intervention: The intervention of interest was the use of SGLT2 inhibitors.
- Comparison: We included studies that utilized any anti-hyperglycemic agent as an active comparator, rather than a placebo.
- Outcome: The primary outcome of interest was the risk of DKA.

The protocol of this systematic review was registered at the International Prospective Register of Systematic Reviews (PROSPERO) (Reference No.: CRD42022385425).

Search strategy

The databases Medline (Ovid) and EMBASE were searched from inception to 19 July 2022. The search strategy was drafted by a librarian and discussed by the research team. Combined, synonymous, and truncated terms of “ketoacidosis” and “sodium-glucose co-transporter 2 inhibitor” were used. Both Medical Subject Headings (MeSH) terms and text words were used. The specific search terms can be found in [Supplementary Table 1](#). The identified articles were exported into Mendeley and duplicates were removed using the publications' unique identifiers, with additional duplicates being identified and eliminated after initial screening. Zotero software was utilized to further check for duplicates and format the bibliography. The titles and abstracts were initially screened by the first author (BA), and if it was not possible to determine inclusion or exclusion based on the criteria from the title and abstract, the full text of the study was reviewed. The resulting full-text studies were reviewed by the first, second, and fourth authors (BA, MAM, and NDM) to identify the finalized dataset of articles. The second and fourth authors (MAM and NDM) made final decisions on the inclusion and exclusion of the studies. The reference lists were checked for additional articles using the Google Scholar search engine.

Eligibility criteria and study selection

For inclusion in the systematic review, the studies had to meet the following criteria: (1) an observational study of adult patients (18 years of age or above) with T2DM taking SGLT2 inhibitors,³⁶ (2) utilize a nonplacebo active comparator, (3) follow an ACNU⁴² or prevalent new user designs,⁴³ (4) have conducted 1:1 propensity score matching,^{44,45} (5) using population-based health database, (6) analyze DKA risk as a primary or secondary outcome or as part of a supplementary, subgroup or sensitivity analysis, and (7) be peer-reviewed. Eligible studies were included regardless of the follow-up duration or date of publication. We did not include non-English studies, letters to journals, opinions, editorials, reviews, and nonobservational studies (e.g., case series).

Data extraction

The data were extracted from the final set of studies onto a standardized template. The key information extracted was study design, country, source of funding (if any), methods (e.g., adjustment for confounders, statistical tests used, and outcome/exposure measurement methods), population characteristics, inclusion and exclusion criteria, duration of follow up, and key findings. The characteristics of the included studies were tabulated using Microsoft Excel™ spreadsheets and grouped by the comparator used to draw direct comparisons. BA created all templates for data extraction, which were then accessed by and made editable for all co-authors (MAM, EM, and NDM) via Google Sheets. Documented weekly and bi-weekly meetings were held among 3 accessors (BA, MAM, and NDM) to track progress, set targets, and resolve discrepancies by consensus.

Quality appraisal

The quality and risk of bias of the included studies were independently assessed by 2 authors (BA and NDM) using the Joanna Briggs Institute (JBI) Critical Appraisal Checklist for Cohort Studies.⁴⁶ Discrepancies were mutually communicated and resolved using email messages and video conference.

Data analysis

Meta-analysis was conducted in the R software (version 4.2.2) (R Foundation for Statistical Computing, Vienna, Austria) by using the *meta* and *metafor* packages. Pooled estimates of the hazard ratio were calculated using the *metagen* function based on the inverse variance method.⁴⁷ The standard error (SE) estimates were calculated based on the 95% confidence interval (95% CI) values. Studies lacking a disclosure of these values and/or hazard ratios were excluded from meta-analysis. Heterogeneity was assessed using Q statistics and Higgins I^2 ,⁴⁸ with values at 25%, 50%, and 75% respectively corresponding to low, medium, and high heterogeneity. The pooled effect size was interpreted based on random-effect models to factor the presence of heterogeneity between the included studies.⁴⁹ ACNU studies were analyzed separately. Meta-regression was performed to identify the source of heterogeneity. Subgroup analyses were carried out to calculate the pooled estimates based on the significant moderator variables identified through meta-regression. Further subgroup analyses were performed stratified by the active comparator and quality score. To evaluate whether the estimated risk was driven by any single study, the leave-one-out analysis was conducted as a sensitivity test. Publication bias was assessed using funnel plot and Egger's test. Statistical significance was considered at P value < 0.05.

Patient and public involvement

No patients or the public were involved in the designing, undertaking, reporting, or dissemination of the protocol of this study.

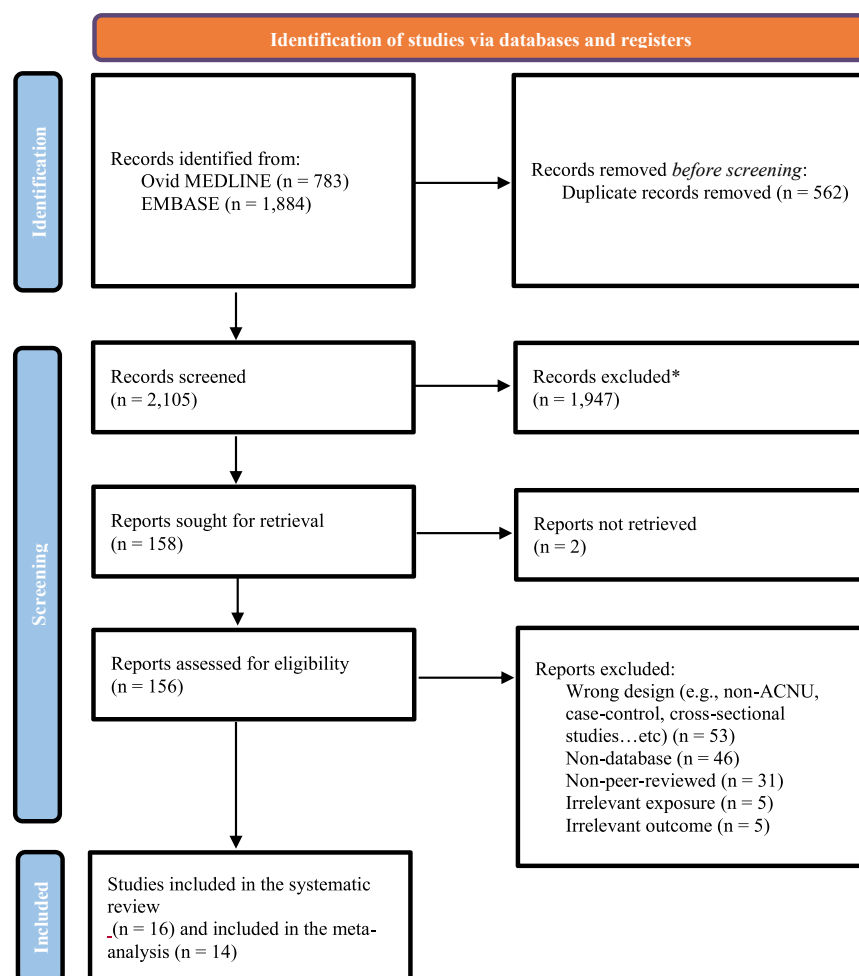


Figure 1. Flow diagram. *These records were excluded based on the title and/or abstract due to being review articles, opinions, case reports, case series, randomized controlled trials (RCTs), non-English publications, animal or *in vitro* experimental reports, and/or studies of type 1 diabetic patients.

Results

Study selection

As shown in [Figure 1](#), our searches retrieved a total of 2,667 records. After removing duplicates, the titles and abstracts of 2,105 studies were screened to eliminate irrelevant records. Full-text evaluations were conducted for the remaining 158 studies against inclusion/exclusion criteria. We excluded 142 studies due to factors such as the study design ($n=53$), absence of database use ($n=46$), lack of peer-review ($n=31$), irrelevant exposure or outcome ($n=10$), or lack of access to the full text ($n=2$).

A summary of the 16 included studies is presented in [Supplementary Table 2](#). Two European studies^{50,51} were included where the study populations were not clearly identified as T2DM patients. This was because the studies made use of national databases and predated the European Commission's authorization for SGLT2 inhibitors' use in T1DM⁵² and in patients without diabetes.^{53,54} We also included a recent Italian study⁵⁵ which estimated that 95% of its cohort had T2DM and excluded chronic insulin users from its analysis. The

cohorts reported in the studies varied greatly due to diverse inclusion and exclusion criteria. These details can be found in [Table 1](#).

Characteristics of included studies

The studies collectively span from January 2010⁵⁵ to December 2019,⁵⁶ and were published between 2017 and 2022, with the majority (10 studies) published after 2020.^{55–64} Seven studies used cohorts from the United States, while the remaining studies were conducted in Canada ($n=2$),^{60,65} Scandinavia ($n=2$),^{50,51} Italy ($n=2$),^{55,61} South Korea ($n=2$),^{64,66} and Singapore ($n=1$)⁵⁸, as shown in [Table 2](#). The majority of the studies were funded by government grants, private foundations, and academic institutions ($n=11$). Three studies received funding from the industry,^{57,67,68} while 2 studies did not disclose any funding sources.^{58,64} All of the eligible studies, except one which specifically examined empagliflozin,⁵⁷ included patients taking any SGLT2 inhibitor in the analysis of DKA risk. The cohorts included in the studies ranged in size from 6,432⁶¹ to 417,514⁶⁵ with a mean age between 53 years⁶⁶ and 72 years^{60,63,64} of both sexes, divided

Table 1
Methodologies of included studies

Study	Study design	Analysis principle	Inclusion criteria	Exclusion criteria
Wang et al. ⁶⁸	Retrospective cohort, ACNU design	Primary analysis: ITT Secondary analysis: PP	T2DM patients who started treatment with SGLT2i, GLP-1RA, DPP4i, SU, TZD, insulin, or other antihyperglycemic agents (except metformin)	(1) Treatment group's prior use of SGLT2i and control groups' prior use of GLP-1RA, DPP4i, SU, TZD, insulin, or other antihyperglycemic agents at any time in the database; (2) patients with less than 365 d of health coverage prior to cohort entry; (3a) type 1 or secondary diabetes at any time prior to or on the index exposure date (broad definition); (3b) type 1 or secondary diabetes diagnosis anytime throughout enrolment (narrow definition); (4) insulin mono-therapy any time prior to the exposure index date (narrow definition); (5) aged less than 40 on index date (narrow definition); (6) having DKA prior to the index date
Ueda et al. ⁵¹	Retrospective cohort, ACNU design	Primary analysis: PP Secondary analysis: ITT	Adults aged 35 or more who started treatment with either SGLT2i or GLP1-RA	(1) Aged less than 35 y; (2) treatment group's prior use of SGLT2i and control group's prior use of GLP-1RA in the past year; (3) Dialysis or renal transplantation at any time before index date; (4) end stage illness (severe malnutrition, cachexia, dementia, coma) at any time before index date (i.e., 10 y lookback in Danish and Swedish datasets, 5 y lookback in Norwegian dataset); (5) drug misuse within the last year; (6) major pancreatic disease (chronic pancreatitis [defined by pancreatic enzyme substitution prescription within last year or diagnosis at any time before index date], pancreatic cancer, major pancreatic surgery at any time before index date); (7) No prescription drug or register entry in the national patient registers within 1 y before index date; and (8) any-cause hospital admission within 30 d before index date.
Pasternak et al. ⁵⁰	Retrospective cohort, ACNU design	Primary analysis: ITT Secondary analysis: AT (AT used solely for DKA risk analysis)	Adults aged 35-84 y who started treatment with SGLT2i or DPP4i	(1) Aged less than 35 y or more than 84 y; (2) treatment group's prior use of SGLT2i and control group's prior use of DPP4i in the past 2 years; (3) Dialysis or renal transplantation at any time before index date; (4) end-stage illness (severe malnutrition, cachexia, dementia, coma) at any time before index date; (5) drug misuse within the past year; (6) major pancreatic disease in the past year; (7) No prescription drug or register entry in the national patient registers within 1 y before index date; and (8) any-cause hospital admission within 30 d before index date.
Fralick et al. ⁵⁹	Retrospective cohort, ACNU design	Primary analysis: PP Secondary analysis: ITT	T2DM patients aged 18 y or more who started treatment with either SGLT2i or metformin	(1) Aged less than 18 y; (2) taking any medication for diabetes in the past 365 d; (3) taking SGLT2i concomitantly with metformin; (4) patients with less than 365 d of health coverage prior to cohort entry; (5) type 1 diabetes; (6) Dialysis; (7) cancer
Kim et al. ⁶⁶	Retrospective cohort, ACNU design	PP	T2DM aged 19 y or more who started treatment with either SGLT2i or DPP4i	(1) Aged less than 19 y; (2) treatment group's prior use of SGLT2i and control group's prior use of DPP4i in the past year; (3) diagnosis with type 1 diabetes mellitus; (2) Malignancy; (3) end-stage renal disease; (4) AIDS; (5) having DKA in the year prior to the index date
Patorno et al. ⁵⁷	Retrospective cohort, ACNU design	AT	T2DM patients aged 18 or more who started treatment with either empagliflozin or DPP4i	(1) Aged less than 18; (2) treatment group's prior use of SGLT2i and control group's prior use of DPP4i in the past year; (3) patients with less than 365 d of health coverage prior to cohort entry; (4) type 1 diabetes, secondary diabetes or gestational diabetes; (5) end-stage renal disease; (6) HIV; (7) organ transplant; (8) Malignancy; (9) Nursing-home admission; (10) exposure to GLP-1RA

(continued on next page)

Table 1 (continued)

Study	Study design	Analysis principle	Inclusion criteria	Exclusion criteria
Han et al. ⁶⁴	Retrospective cohort, ACNU design	ITT	T2DM patients aged 65 y or more who started treatment with either SGLT2i or DPP4i	(1) Aged less than 65; (2) treatment group's prior use of SGLT2i and control group's prior use of DPP4i in the past year; (3) diagnosis with type 1 diabetes or gestational diabetes
Fralick et al. ⁶⁰	Retrospective cohort, ACNU design	Primary analysis: PP Secondary analysis: ITT	T2DM patients aged 66 y or more who started treatment with either SGLT2i or DPP4i	(1) Aged less than 66 y; (2) treatment group's prior use of SGLT2i and control group's prior use of DPP4i in the past year; (3) patients with less than 365 d of health coverage prior to cohort entry; (4) Invalid health card number; (5) type 1 diabetes; (6) Dialysis; (7) cancer
Wang et al. ⁶⁷	Retrospective cohort, ACNU design	Primary analysis: ITT Secondary analysis: PP	T2DM patients who started treatment with any of SGLT2i, GLP-1RA, DPP4i, metformin, SU, nateglinide, repaglinide, TZD, acarbose, insulin, or miglitol	(1) Treatment group's prior use of SGLT2i and control groups' prior use of GLP-1RA, DPP4i, metformin, SU, nateglinide, repaglinide, TZD, acarbose, insulin, or miglitol at any time in the database; (2) patients with less than 365 d of health coverage prior to cohort entry; (3a) type 1 or secondary diabetes at any time prior to or on the index exposure date (broad definition); (3b) type 1 or secondary diabetes diagnosis anytime throughout all enrolment records (narrow definition); (4) insulin mono-therapy any time prior to the exposure index date (narrow definition); (5) aged less than 40 on index date (narrow definition);
Patorno et al. ⁶³	Retrospective cohort, ACNU design	Primary analysis: AT secondary analysis: ITT	T2DM patients aged 66 y or more who started treatment with either SGLT2i or GLP-1RA	(1) Aged less than 66 y; (2) treatment group's prior use of SGLT2i and control group's prior use of GLP-1RA in the past year; (3) patients with less than 365 d of health coverage prior to cohort entry; (4) type 1 diabetes or secondary diabetes; (5) Malignancy; (6) end-stage renal disease; (7) HIV; (8) organ transplant; (9) Nursing home admission; (10) Hospitalization for acute myocardial infarction, coronary revascularization, unstable angina, ischemic or hemorrhagic stroke, a transient ischemic attack, and heart failure in the 60 d prior to cohort entry
Douros et al. ⁶⁵	Retrospective cohort, prevalent new user design	AT	T2DM patients aged 19 y or more who started treatment with either SGLT2i or DPP4i	(1) Aged <18 y (<19 y in Alberta and <66 y in Ontario because Ontario site did not include younger patients); (2) treatment group's prior use of SGLT2i and control group's prior use of DPP4i in the past year; (3) patients with less than 365 d of health coverage prior to cohort entry; (4) taking SGLT2i and DPP4is concomitantly; (5) patients who switched to DPP4is after being treated with SGLT2is; (6) having DKA in the year prior to cohort entry
Goh et al. ⁵⁸	Retrospective cohort, ACNU design	ITT	T2DM patients aged 30 or more who started treatment with either SGLT2i or DPP4i	(1) Aged less than 30 y; (2) treatment group's prior use of SGLT2i and control group's prior use of DPP4i in the past year; (3) aged less than 40 y on insulin monotherapy; (4) nonresidents without regular management in Singapore; (5) records which did not show the age and gender; (6) patients with missing HbA1c results during the follow-up period; (7) a death record within 30 d of treatment initiation
Patorno et al. ⁶²	Retrospective cohort, ACNU design	Primary analysis: AT secondary analysis: ITT	T2DM patients aged 18 or more who started treatment with either SGLT2i or GLP-1RA	(1) Aged less than 18 (Optum and MarketScan) and less than 66 in Medicare; (2) treatment group's prior use of SGLT2i and control group's prior use of GLP-1RA in the past year; (3) patients with less than 365 d of health coverage prior to cohort entry; (4) type 1 diabetes or secondary diabetes; (5) cancer; (6) end-stage renal disease or renal replacement therapy; (7) HIV; (8) Nursing-home admission; (9) hospitalization for acute MI, coronary revascularization, unstable angina, ischemic or hemorrhagic stroke, transient ischemic attack, and heart failure in the 60 d before cohort entry

Table 1 (continued)

Study	Study design	Analysis principle	Inclusion criteria	Exclusion criteria
Dawwas et al. ⁵⁶	Retrospective cohort, ACNU design	PP	T2DM patients aged more than 18 y who started treatment with any of SGLT2i, DPP4i, or SU	(1) Aged 18 y or less; (2) treatment group's prior use of SGLT2i and control groups' prior use of DPP4i or SU; (3) patients with less than 365 d of health coverage prior to cohort entry; (4) diagnosis with type 1 diabetes; (5) taking SGLT2i and DPP4i or sulfonylureas concomitantly; (6) end-stage renal disease; (7) having DKA
Longato et al. ⁶¹	Retrospective cohort, ACNU design	Primary analysis: ITT Secondary analysis: As treated	T2DM patients who are residents of Veneto and started treatment with either SGLT2i or DPP4i	(1) Residing in Veneto for less than 1 y; (2) treatment group's prior use of SGLT2i and control group's prior use of DPP4i in the past 7 m; (3) patients who switched to DPP4is after being treated with SGLT2is, and vice versa; (4) taking bolus insulin; (5) kidney disease in the past year
Baviera et al. ⁵⁵	Retrospective cohort, ACNU design	ITT	Adults aged 50 or more who started treatment with any of SGLT2i, GLP-1RA, DPP4i, metformin, SU, meglitinides, TZD, or acarbose	(1) Aged less than 50 y; (2) treatment groups' prior use of GLP-1RA or SGLT2i, and control groups' prior use of DPP4i, metformin, SU, meglitinides, TZD, or acarbose in the past 5 y; (3) new users of insulin; (4) taking less than 2 packages of the antihyperglycemic agent of interest per year

Abbreviations used: ACNU, active comparator new user; AT, as-treated exposure definition; CVD, cardiovascular disease; DKA, diabetic ketoacidosis; DPP4i, dipeptidyl peptidase 4 inhibitors; GLP-1RA, glucagon-like peptide 1 receptor agonists; HIV, human immunodeficiency virus; ITT, intent-to-treat exposure definition; MI, myocardial infarction; PP, per-protocol exposure definition; SGLT2i, sodium-glucose transporter-2 inhibitors; SU, sulfonylureas; TZD, thiazolidinedione; T2DM, type 2 diabetes mellitus.

equally into new users of SGLT2 inhibitors and new users of an active comparator(s). Male patients were 46%⁵⁹ and up to 67%⁶¹ of the participants in almost all of the studies. The sex composition was not reported in 2 studies.^{67,68} The median follow-up ranged from 4.31⁵⁶ to 21.6 months.⁵⁵ Five studies excluded patients with DKA at the baseline.^{56,65-68}

Dipeptidyl peptidase-4 (DPP4) inhibitors were the most common comparators, used in 11 (68.75%) studies.^{50,55-58,60,61,64-67} while glucagon-like peptide 1 receptor agonists (GLP-1RA) were the second most commonly used class (25%).^{51,62,63,67} Other comparators included sulfonylureas (n=2),^{56,67} metformin (n=2),^{59,67} thiazolidinediones (n=1), insulin (n=1),⁶⁷ and any other non-SGLT2-inhibitor antihyperglycemic agents (n=1).⁶⁸ Ten studies reported that patients on SGLT2 inhibitors had a greater chance of experiencing DKA events (versus control),^{50,51,56,57,59,60,62,63,65,67} while 6 studies concluded that SGLT2 inhibitors had a neutral association with DKA incidence.^{55,58,61,64,66,68} The crude number of DKA events was reported in all except one study⁶⁴ and ranged from one event observed in SGLT2-inhibitor users in a small study⁶¹ to 499 events in a large study.⁶² Overall, 42.77% (n=2348) of the observed DKA events occurred in the comparator groups, with events ranging from none in a small study⁶¹ to 287 in a large study.⁶² The absolute rate of DKA events was low and ranged from 0.6⁶⁶ to 6.3⁵⁶ per 1000 person-years in new users of SGLT2 inhibitors, and 0.6^{50,51} to 4.5⁵⁶ per 1000 person-years in new users of active comparators.

Most of the studies (n=15) adopted an ACNU design and only one study utilized the prevalent new user design.⁶⁵ Six studies assessed the risk of DKA using an intention-to-treat (ITT) principle in the primary analysis,^{55,58,61,64,67,68} while the remaining studies adjusted the follow-up to observe their cohorts per-protocol (PP) or as-treated (AT). Where a narrow definition of T2DM was adopted versus a broad one,^{67,68} we

only considered evidence that favored the exclusion of T1DM and secondary diabetes during follow-up (i.e., the narrow definition).

Quality of evidence

The overall quality of the included studies is presented in [Supplementary Table 3](#). All of the studies met 7 or more of the appraisal criteria. However, 12 studies had a lower risk of bias as they satisfied more than 80% of the criteria, while 4 studies received a “no” or “unclear” answer to 3 or 4 questions in the appraisal checklist, representing a higher risk of bias.^{55,58,67,68} Confounder choice and matching were clearly reported in all studies except two.^{67,68} It was not clear whether the participants were free of DKA upon exposure initiation, or whether follow-up was complete, in the majority of the included studies. Statistical analyses performed in three studies was found to be suboptimal or undermined by faulty design aspects, including a failure to adjust for key variables.^{55,67,68}

Meta-analysis

Two studies which initially satisfied our inclusion criteria^{55,61} were excluded from meta-analysis because they observed no DKA events in their comparator arms, resulting in undefined median survival times. Fourteen studies were deemed to have sufficient data to perform meta-analysis,^{50,51,56-60,62-68} of which 3 studies derived evidence from multiple cohorts.^{56,62,67} A forest plot was generated as shown in [Figure 2](#). The analysis of observational studies indicates a 33% higher real-world risk of DKA among T2DM patients taking SGLT2 inhibitors (versus non-SGLT2-inhibitor antihyperglycemic agents) (pooled hazard ratio (HR) = 1.33, 95% confidence interval (95% CI): 1.14-1.55, $P < 0.01$). The analysis of ACNU observational studies alone suggests a comparable 27%

Table 2
Characteristics of included studies

Author	Country	Enrollment	Intervention	Comparator	Follow-up (month)	Study population (values after PSM, for SGLT2 users)					Outcomes measured
						Age (years)	Male (%)	Ethnicity	Comorbidities	Baseline medications	
Wang et al. ⁶⁸	USA	2013 - 2014	SGLT2i class	Non-SGLT2i antihyperglycemic agents	Not reported	55.0 ± 6.6	Not reported	Not available	Not available	Metformin (91%), SU (55%), DPP4i (52%), TZD (29%), insulin (10%), GLP-1RA (8%), other antihyperglycemic agents (5%)	Hospitalization or emergency room visit due to DKA
Ueda et al. ⁵¹	Denmark and Sweden	2013 - 2016	Dapagliflozin (61%), empagliflozin (38%), canagliflozin (1%)	GLP-1RA	8.88 (median)	61.0 ± 10.0	61	Not available	Diabetic, other complications (19%), other IHD (16%), obesity (15%), diabetic eye complications (13%), psychiatric disorder (10%), ACS (8%), atrial fibrillation (7%), other lung diseases (7%), cancer (7%), HF/ cardiomyopathy (6%), arterial disease (6%), renal disease (4%), stroke (4%), other cerebrovascular disease (4%), other arrhythmia (4%), COPD (3%), rheumatic disease (3%), VTE (3%), valve disorders (2%), liver disease (2%), fracture in the past year (2%), serious UTI (2%), coronary revascularization in past year (1%), acute pancreatitis (1%), other cardiac surgery or invasive procedure in past year (<1%), lower limb amputation (<1%), DKA (<1%)	Metformin (80%), DPP4i (35%), insulin, intermediate or long acting (29%), insulin, fast acting (16%), SU (23%), GLP1 agonists (10%), other diabetes drugs (4%)Lipid-lowering drugs (71%), ARB/ACE-I (69%), b-blockers (36%), platelet inhibitors (34%), CCB (33%), NSAIDs (23%), opioid (17%), other diuretic (18%), Anxiolytic, hypnotic, or sedative (16%), antidepressant (17%), loop diuretic (13%), B2 agonist inhalant (9%), glucocorticoid inhalant (9%), nitrate (7%), anticoagulant (7%), OCS (7%), antipsychotics (4%), Anticholinergic inhalant (3%), digoxin (2%)	Primary outcomes: Lower limb amputation, bone fracture, DKA, AKI, serious UTI, VTE, and acute pancreatitis. Secondary outcomes: Toe/ metatarsal amputation and major osteoporotic fracture.
Pasternak et al. ⁵⁰	Denmark, Norway and Sweden	2013 - 2016	Dapagliflozin (83%), empagliflozin (16%), canagliflozin (1%)	DPP4i	8.40 (median)	61.0 ± 10.0	60	Not available	Diabetic complications (29%), other IHD (17%), psychiatric disorder (10%), ACS (7%), atrial fibrillation (7%), other lung diseases (7%), cancer (7%), HF/ cardiomyopathy (6%), arterial disease (including amputations) (6%), renal disease (5%), stroke (4%), other cerebrovascular disease (4%), other arrhythmia (4%), COPD (4%), rheumatic disease (3%), valve disorders (2%), VTE (2%), liver disease (2%), fracture in the past year (2%), coronary revascularization in past year (1%), other cardiac surgery or invasive procedure in past year (1%)	Metformin (79%), insulin (32%), SU (21%), GLP1 agonists (10%), other diabetes drugs (3%)Lipid-lowering drugs (67%), ARB/ACE-I (66%), β-blockers (36%), platelet inhibitors (36%), CCB (30%), NSAID (26%), opioid (20%), other diuretic (18%), Anxiolytic, hypnotic, or sedative (18%), antidepressant (16%), loop diuretic (14%), B2 agonist inhalant (10%), glucocorticoid inhalant (10%), nitrate (7%), anticoagulant (7%), OCS (7%), antipsychotics (4%), Anticholinergic inhalant (3%), digoxin (2%)	Primary outcomes: Major cardiovascular events and HF. Secondary outcomes: Individual components of the cardiovascular composite and any cause death. Supplementary outcomes: Lower-limb amputation and DKA

Fralick et al. ⁵⁹	USA	2013 - 2018	SGLT2i class	Metformin	4.83 (median)	53.5 ± 10.0	48	Not available	Hypertension (71%), dyslipidemia (68%), obese or overweight (33%), IHD (11%), diabetic neuropathy (8%), smoking (7%), HF (3%), chronic kidney injury (3%), diabetic retinopathy (3%), diabetic nephropathy (3%), genital infection (3%), ischemic stroke or TIA (2%), AKI (1%), hypoglycemia (0.4%), DKA (0.2%)	Statin (38%), ACE inhibitor (24%), angiotensin receptor blocker (20%), B-blocker (15%), Diuretic (14%), CCB (10%), antiplatelet (5%)	Primary outcome: a Composite of HF, MI or stroke Safety outcomes: Hypoglycemia, DKA, genital infection, lactic acidosis and AKI
Kim et al. ⁶⁶	South Korea	2013 - 2017	SGLT2i class	DPP4i	10.49 (mean)	53.1 ± 11.8	55	Not available	Dyslipidemia (82%), hypertension (60%), respiratory tract infection (21%), other IHD (14%), retinopathy (12%), other heart disease (12%), asthma (12%), nephropathy (10%), CKD (except ESRD) (7%), neuropathy (7%), osteoporosis (7%), cerebrovascular event (5%), COPD (5%), alcohol use (5%), stroke (4%), connective tissue disease (3%), UTI (4%), HF (4%), hypoglycemia (2%), gastrointestinal infection (2%), atrial fibrillation (2%), acute MI (1%), pancreatitis (1%), peripheral arterial occlusive disease (<1%), tobacco use (<0.5%), obesity (<0.5%)	Metformin (81%), SU (36%), insulin (13%), TZD (9%), a-glucosidase inhibitor (5%), meglitinide (<1%) Statins (53%), angiotensin II receptor antagonist (47%), steroids (35%), CCB (30%), thiazide diuretics (15%), fibrate (11%), other anti-hypertensive agents (10%), ezetimibe (9%), loop diuretics (4%), ACE inhibitors (3%), other diuretics (2%)	Hospitalization for DKA
Patorno et al. ⁵⁷	USA	2014 - 2017	Empagliflozin	DPP4i	6.00 (mean)	60.2 ± 9.1	55	84% white	Hyperlipidemia (80%), hypertension (78%), CVD (28%), IHD (21%), diabetic neuropathy (18%), diabetic nephropathy (10%), chronic kidney disease (9%), Oedema (8%), Hypoglycemia (8%), COPD (7%), HF (6%), atrial fibrillation (6%), ischemic stroke (6%), PAD or surgery (6%), diabetic retinopathy (6%), diabetes with peripheral circulatory disorders (5%), osteoporosis (4%), fractures (3%), falls (3%), AKI (2%), dementia (2%), TIA (2%), other cerebrovascular disease (2%), acute MI (2%), diabetic foot (2%), coronary revascularization (1%), lower-limb amputations (0.4%), DKA (0.1%)	Metformin (62%), SU (27%), insulin (18%), GLP-1RAs (10%), glitazones (6%), Statins (70%), ACE inhibitors (46%), Beta-blockers (35%), angiotensin II receptor blockers (31%), opioids (31%), CCB (25%), OCS (24%), other lipid-lowering drugs (14%), thiazides (13%), benzodiazepine (13%), antiplatelet agents (12%), loop diuretics (11%), oral anticoagulants (6%), bisphosphonates (1%)	Primary outcomes: Composite of MI/stroke, and hospitalization for HF Safety outcomes: bone fractures, lower-limb amputations, hospitalization for DKA, and AKI

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Table 2 (continued)

Author	Country	Enrollment	Intervention	Comparator	Follow-up (month)	Study population (values after PSM, for SGLT2 users)					Outcomes measured
						Age (years)	Male (%)	Ethnicity	Comorbidities	Baseline medications	
Han et al. ⁶⁴	South Korea	2014 - 2016	Dapagliflozin, empagliflozin, and ipragliflozin	DPP4i	12.65 (mean)	71.9 ± 5.5	42	Not available	CVD (46%), diabetic retinopathy (34%), diabetic neuropathy (31%), angina pectoris (26%), stroke (20%), diabetic nephropathy (18%), HF (11%), cancer (15%), unstable angina (9%), atrial fibrillation (5%), MI (4%), severe hypoglycemia (4%), percutaneous coronary intervention with stent (3%), CKD (3%), DKA/lactate acidosis (<1%), PAD (<0.5%)	Metformin (71%), SU (52%), insulin (18%), TZD (13%), α-glucosidase (9%), meglitinide (1.5%), GLP-1RA (0.1%), Statin (67%), angiotensin II receptor blockers (60%), low dose acetylsalicylic acid (38%), dihydropyridines (31%), B-Blockers (26%), receptor P2Y12 antagonists (17%), low-ceiling diuretics (14%), high-ceiling diuretics (11%), ACE inhibitors (5%), non-hydropyridine (5%), Aldosterone antagonists (5%), warfarin (2%)	Hospitalization for HF, all-cause death, MI, stroke, DKA, bone fracture, severe hypoglycemia, genital infection, and UTI
Fralick et al. ⁶⁰	Canada	2015 - 2019	SGLT2i class	DPP4i	13.00 (mean)	72	60	Not available	Hypertension (39%), coronary artery disease (20%), cataract (19%), glaucoma (12%), heart failure (7%), CKD (6%), atrial fibrillation (3%), CVD (3%), PAD (3%), COPD (3%), obesity (3%), dementia (3%), nephropathy (2%), retinopathy (1%), neuropathy (1%), Hypoglycemia (1%), cirrhosis (1%), DKA (0.2%)	Metformin (81%), insulin (30%), SU (30%), TZD (0.6%), meglitinide (0%), Statins (82%), ACE inhibitors (44%), Beta-blockers (38%), calcium-blockers (37%), Diuretics (32%), angiotensin-receptor blocker (27%), antiplatelets (12%), anticoagulants (12%), glucocorticoids (7%), spironolactone (4%), digoxin (2%), dementia medications (1%), sacubitril and valsartan (0.4%)	Primary outcome: a Composite of HF hospitalization and all-cause mortality Secondary outcomes: Cardiovascular mortality, stroke, HF, DKA and hypoglycemia
Wang et al. ⁶⁷	USA	2013 to last data availability ²	SGLT2i class	DPP4i, GLP-1RA, SU, metformin, TZD, insulin,	Not reported	Not reported	Not reported	Not available	Not available	Not available	Hospitalization or emergency room visit due to DKA
Paterno et al. ⁶³	USA	2013 - 2016	Canagliflozin (76.9%), dapagliflozin (13.1%), and empagliflozin (11.1%)	GLP-1RA	5.50 (median)	71.6 ± 5.0	46	83% white, 8% Black, 3% Hispanic, 2% Asian, and 3% unidentified	Hypertension (92%), hyperlipidemia (90%), CVD (45%), ischemic heart disease (33%), osteoarthritis (32%), diabetic neuropathy (26%), UTI (17%), CKD (16%), obstructive sleep apnea (14%), oedema (14%), PAD or surgery (13%), COPD (13%), diabetic nephropathy (11%), diabetic retinopathy (11%), HF (11%), atrial fibrillation (11%), ischemic/hemorrhagic stroke (11%), CKD, stage 3+ (11%), osteoporosis (9%), peripheral circulatory disorders (8%),	Metformin (58%), SU (37%), insulin (24%), DPP4i (24%), glitazones (8%), Statins (77%), B-blockers (49%), ACE inhibitors (47%), angiotensin II receptor blockers (37%), opioids (36%), CCB (34%), UTI antibiotics (30%), OCS (28%), loop diuretics (20%), antiplatelet agents (17%), thiazide diuretics (16.5%), benzodiazepines (15%), nitrates (9%), anticoagulants (9%), other diuretics (4%),	Primary outcomes: MACE (i.e., MI, stroke, or cardiovascular death) and hospitalization for HF Secondary outcomes: DKA, genital infections, fractures, lower-limb amputations, AKI, severe UTI, and overall mortality

									hypoglycemia (5%), fractures (5%), falls (5%), dementia (5%), pneumonia (4%), AKI (4%), diabetic foot (3%), TIA (3%), genital infections (3%), acute MI (2%), coronary revascularization (<1%), lower-extremity amputation (0.5%), DKA (<0.5%)	bisphosphonates (4%), digoxin (3%)	
Douros et al. ⁶⁵	Canada & UK	2013 to last data availability ¹	Canagliflozin (42.3%), dapagliflozin (30.7%), and empagliflozin (27%)	DPP4i	10.80 (mean)	63.8 ± 9.5	59	Not available	Cancer (excluding nonmelanoma skin cancer) (10.5%), other kidney disease (5%), diabetic nephropathy (3.6%), MI (2.6%), diabetic retinopathy (2.6%), PAD (2.3%), diabetic neuropathy (1.9%), alcohol-related disorders (1.7%), ischemic stroke (1.2%), dialysis (<0.5%)	Metformin (87%), SU (52%), insulin (28%), GLP-1RA (4%), TZD (3%), meglitinides (2%), α-glucosidase inhibitors (2%) Statins (77%), ACE inhibitors (46%), angiotensin II receptor blockers (32%), CCB (31%), β-blockers (28%), thiazide diuretics (22%), NSAIDs (19%), acetylsalicylic acid (18%), other lipid-lowering therapy (11%), loop diuretics (10%), other diuretics (9%), other antiplatelet agents (7%), oral anticoagulants (7%), OCS (6%), atypical antipsychotics (4%), aldosterone antagonists (3%), digitalis-like agents (1%), direct renin inhibitors (<0.1%)	Hospitalization or emergency department visit for DKA
Goh et al. ⁵⁸	Singapore	2015 - 2018	Dapagliflozin, empagliflozin, and canagliflozin	DPP4i	Not reported	56.3 ± 10.2	55	Chinese (61%), Indian (17%), Malay (14%) and others (9%)	CVD (30%), hypertension (25%), hyperlipidemia (25%), insulin resistance (15%), Poor control (6%), DM kidney complications (2%), retinopathy (3%), cancer (1%), hypoglycemia (1%), foot ulcer (1%), neuropathy (0.5%), peripheral angiopathy (0.6%)	Metformin (64%), SU (56%), insulin (21%), acarbose (4%)	Glycemic control, and any-cause/cause-specific hospitalizations (DKA included), and all-cause death
Patorno et al. ⁶²	USA	2013 - 2017	Canagliflozin (59.3%), empagliflozin (22.3%), and dapagliflozin (18.3%)	GLP-1RA	7.00 (median)	60.7 ± 12.0	49	75.2% White, 9.6% black, 8% Hispanic, and 2.8% Asian, with 4.5% unknown	Hyperlipidemia (80%), hypertension (78%), obesity (36%), hyperglycemia (32%), CVD (28%), hypothyroidism (20%), diabetic neuropathy (19%), other coronary atherosclerosis (19%), osteoarthritis (19%), obstructive sleep apnea (18%), anxiety or sleep disorder (17%), depression (14%), UTI (12%), diabetic nephropathy (10%), CKD (10%), diabetic retinopathy (9%), oedema (9%), diabetes with other ophthalmic manifestations (8%), conduction disorders	Metformin (61%), SU (31%), DPP4i (21%), insulin (21%), glitazones (7%), other antidiabetic agent (2.5%) ACEs or ARBs (75%), statins (70%), opioids (36%), B-blockers (35%), antidepressants (30%), NSAIDs (28%), CCB (26%), UTI antibiotics (23%), other lipid-lowering drugs (18%), COPD or asthma medications (18%), OCS (17%), gabapentinoids (15%), thiazide diuretics (14%), benzodiazepines (14%), loop diuretics (12%), antiplatelet agents (9%), other	Primary outcomes: MI or stroke hospitalization, and hospitalization for HF Secondary outcomes: MI, ischemic or hemorrhagic stroke; all-cause mortality; and a composite outcome of MI, ischemic or hemorrhagic stroke, or all-cause mortality. DKA assessed in the sensitivity analysis.

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Author	Country	Enrollment	Intervention	Comparator	Follow-up (month)	Study population (values after PSM, for SGLT2 users)					Outcomes measured
						Age (years)	Male (%)	Ethnicity	Comorbidities	Baseline medications	
									(8%), PAD or surgery (8%), COPD (8%), Asthma (8%), hypoglycemia (7%), diabetes with other complication (7%), cardiac valve disorders (7%), other cardiovascular disease (7%), miscellaneous renal disease (7%), other disorders of thyroid gland (7%), HF (6%), atrial fibrillation (6%), ischemic stroke (6%), history of coronary procedure (5%), disorders of electrolyte (5%), non-alcoholic steatohepatitis or fatty liver disease (5%), overweight (5%), infections of lower extremities (4%), erectile dysfunction (4%), stable angina (4%), other cerebrovascular disease (4%), hypertensive nephropathy (4%), proteinuria (4%), kidney or urinary stone (4%), osteoporosis (4%), diabetes with peripheral circulatory disorders (3%), old MI (3%), cardiomyopathy (3%), other cardiac dysrhythmia (3%), falls (3%), disorders of fluid balance (3%), pneumonia (3%), other liver disease (3%), diabetic foot (2%), unstable angina (2%), TIA (2%), AKI (2%), dementia (2%), acute MI (1%), coronary revascularization (1%), fractures (1%), hyperosmolar hyperglycemic nonketotic syndrome (0.7%), DKA (0.4%), lower limb amputation (0.3%)	antihypertensives (8%), other anxiolytics or hypnotics (8%), nitrates (6%), oral anticoagulants (6%), mineralocorticoid receptor antagonists (4%), antiparkinsonian medications (3%), osteoporosis medications (2%), antipsychotics (2%), digoxin (1%), antiarrhythmics (1%), dementia medications (1%)	
Dawwas et al. ⁵⁶	USA	2013 - 2019	SGLT2i class	DPP4i, SU	4.31 (median)	60.7 ± 11.0	57	66% White, 17% Hispanic, 14% to 13% black, and 4% Asian, with 15% unknown	Hypertension (69%), hyperlipidemia (46%), chronic lung disease (14%), chronic kidney disease (12%), PAD (7%), tobacco use (7%), heart failure (6%), cancer (5%), liver disease (5%), arthritis (4%), depression (4%), venous thromboembolism (2%),	Metformin (65%), insulin (22%), GLP-1RA (9%), TZD (7%), amylin (0.2%) Statins (59%), ACE inhibitors (39%), thiazide diuretics (27%), angiotensin II receptor blocker (24%), b-blockers (23%), CCB (20%), proton pump inhibitor (20%), selective serotonin reuptake	Hospitalization for DKA

									alcohol abuse (1%), drug abuse (1%), stroke (0.1%)	inhibitor (14%), loop diuretics (9%), α -Adrenergic blockers (6%), potassium diuretics (3%), aldosterone antagonists (3%), direct vasodilators (1%)	
Longato et al. ⁶¹	Italy	2011 - 2018	Empagliflozin (52%), dapagliflozin (38%), and canagliflozin (10%)	DPP4i	18.00 (median)	63.4 $\pm$ 8.7	67	Not available	Hypertension (82%), dyslipidemia (73%), chronic pulmonary disease (33%), CVD (21%), ischemic heart disease (17%), prior MI (11%), history of cancer (10%), stroke or TIA (4%), PAD (2%), history of HF (2%), inflammatory disease (2%), history of severe hypoglycemia (0.9%), CKD (0.6%), neuropathy (0.3%), retinopathy (0.2%), diabetic nephropathy (0%)	Metformin (90%), RAS blockers (69%), statins (64%), sulfonylurea (44%), anti-platelet agents (37%), B-blockers (36%), other diuretics (16%), Fibrates or PUFA (12%), pioglitazone (11%), other anti-hypertensives (7%), GLP-1RA (6%), ezetimibe (3%), thiazide diuretics (1.6%)	Primary outcomes: MI, stroke, or all-cause death. Secondary endpoints: Individual MACE components, first hospitalization for HF, arterial revascularizations, and hospital admission for any cardiovascular cause. Secondary analysis: Adverse endpoints of UTI, bone fracture, DKA, acute renal disease, pancreatitis, and pancreatic cancer
Baviera et al. ⁵⁵	Italy	2010 - 2018	SGLT2i class	DPP4i	21.60/19.20 (median) $\pm$ 7.4	68.6 $\pm$ 7.5/67.5	62/54	Not available	Cardiovascular disease (15%), cancer (6.6%), CVD (4.4%), PAD (3.9%), HF (3.7%), COPD (2.2%), neuropathy (1.4%), renal disease (1.2%), lower limb complication (0.9%), diabetic retinopathy (0.2%)	Metformin (92%), SU (56%), insulin (38%), DPP4i (31%), glitazones (22%), glinides (12%), acarbose (10%), GLP-1RA (1.4%) Antihypertensive drugs (83%), ACE-I/ARBs (72%), lipid lowering drugs (70%), antiplatelet drugs (38%), oral anticoagulant drugs (6%)	Death from any cause, hospital admission for cerebrovascular disease, CVD, ischemic stroke, ACS, HF, PAD, and lower limb complications, serious adverse events, hospital admission for hypoglycemia, DKA, diabetes with coma, amputation, acute renal failure, syncope, and fracture as primary diagnosis.

Abbreviations used: ACE, angiotensin-converting enzyme; ACS, acute coronary syndrome; AKI, acute kidney injury; CCB, calcium channel blocker; CKD, chronic kidney disease; COPD, chronic obstructive pulmonary disease; CVD, cardiovascular disease; DKA, diabetic ketoacidosis; DPP4i: dipeptidyl peptidase 4 inhibitors; ESRD, end-stage renal disease; GLP-1RA: glucagon-like peptide 1 receptor agonists; HF, heart failure; IHD, ischemic heart disease; MACE, major adverse cardiovascular events; MI, myocardial infarction; NSAIDs, nonsteroidal anti-inflammatory drugs; OCS, oral corticosteroids; PAD, peripheral arterial disease or peripheral vascular disease; SGLT2i, sodium-glucose transporter-2 inhibitors; SU, sulfonylureas; TIA, transient ischemic attack; TZD, thiazolidinedione; UTI: urinary tract infection; VTE, venous thromboembolism.

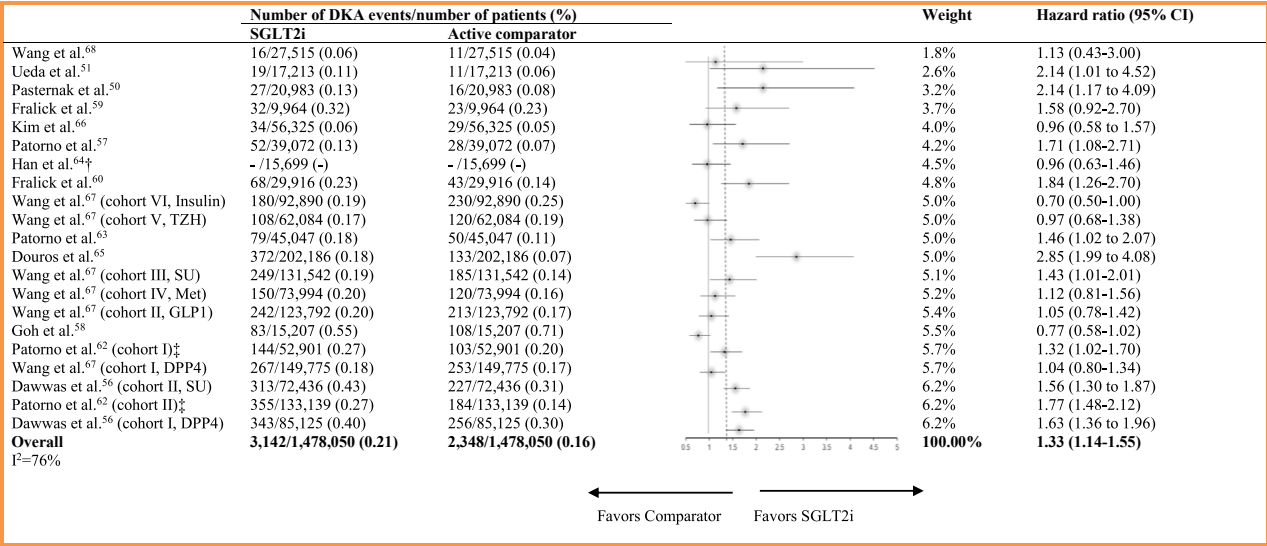


Figure 2. Forest plot. DPP4: dipeptidyl peptidase 4 inhibitors; GLP1, glucagon-like peptide 1 receptor agonists; SU, sulfonylureas; DKA, diabetic ketoacidosis; Met, Metformin; Thiazolidinedione, TZH. †The crude number of DKA events was not reported. ‡Cohort I: with baseline cardiovascular disease (CVD); Cohort II: without baseline CVD.

significant increase in DKA risk in patients treated with SGLT2 inhibitors (versus. non-SGLT2-inhibitor antihyperglycemic agents) (pooled HR = 1.27, 95% CI: 1.10-1.47, $P < 0.01$).

Subgroup analyses

In sub-analysis that examined studies based on the class of active comparators used, no significant association was found between the risk of DKA and the use of SGLT2 inhibitors when metformin served as the active comparator (pooled HR = 1.24, 95% CI: 0.91-1.69, $P = 0.17$). SGLT2 inhibitors significantly increased the risk of DKA by 1.40-, 1.43- and 1.53-fold when compared with DPP4 inhibitors (pooled HR = 1.40, 95% CI: 1.05-1.87, $P = 0.02$), GLP-1RA (pooled HR = 1.43, 95% CI: 1.12-1.83, $P < 0.01$) and sulfonylureas (pooled HR = 1.53, 95% CI: 1.30-1.80, $P < 0.001$), respectively. Meta-regression analysis identified the study location ($P = 0.02$), the analysis principle ($P < 0.001$), exclusion of chronic comorbidities ($P = 0.007$), and canagliflozin ($P = 0.04$) as significant moderator variables. Limiting the design to ACNU studies alone, canagliflozin ceased to be a significant moderator (refer to Table 3 for details). Subgroup analyses based on geographical territory indicated a significant ($P < 0.001$) association between SGLT2 inhibitors and DKA for studies conducted in North America (USA and Canada) (pooled HR = 1.32, 95% CI: 1.15-1.53) and Scandinavia (Norway, Sweden and Denmark) (pooled HR = 2.14, 95% CI: 1.32-3.46), but not for those conducted in Asia (South Korea and Singapore) (pooled HR = 0.85, 95% CI: 0.69-1.05). In subgroup analyses based on the concept of analysis, a significant ($P < 0.001$) association between SGLT2 inhibitors and DKA was identified for studies using AT (pooled HR = 1.60, 95% CI: 1.36-1.88) and PP (pooled HR = 1.58, 95% CI: 1.41-1.77), but not ITT as the primary analysis principle (pooled HR=0.99, 95% CI: 0.85-1.14). Canagliflozin was the dominant SGLT2 inhibitor observed in 3 studies in our review.^{62,63,65} Though in ACNU studies meta-regression did not identify this molecule as a significant moderator variable, subgroup analysis

suggested a higher DKA risk when canagliflozin was the dominant SGLT2-inhibiting molecule (pooled HR = 1.54, 95% CI: 1.26-1.88, $P < 0.001$), as compared with studies in which it was not (pooled HR = 1.23, 95% CI: 1.05-1.44, $P < 0.001$). Subgroup analyses based on the exclusion of certain comorbidities identified a significant ($P < 0.001$) association between SGLT2 inhibitors and DKA for studies ($n = 9$) that excluded patients with certain comorbidities (cancer, kidney disease, human immunodeficiency virus (HIV), organ transplant, and end-stage illness) (pooled HR = 1.60, 95% CI: 1.47-1.74), but not for studies that did not make such exclusions (pooled HR = 0.99, 95% CI: 0.85-1.14).

Sensitivity analyses

Excluding studies with lower-quality scores^{58,67,68} yielded a higher real-world significant DKA risk estimate in patients receiving SGLT2 inhibitors (versus. non-SGLT2-inhibitor antihyperglycemic agents) (pooled HR = 1.60, 95% CI: 1.39-1.85, $P < 0.01$). The analysis identified 3 cohorts from 2 studies^{56,62} as having the highest statistical weight. As summarized in Table 4, sensitivity analysis showed that the association between SGLT2 inhibitors (versus. non-SGLT2-inhibitor antihyperglycemic agents) and DKA remained significantly positive and was not dictated by any single study. To obtain the most conservative pooled risk estimate, we limited the analysis to ACNU studies and excluded those cohorts^{56,58,62,67,68} that were most likely to skew the estimation of the overall effect size—from studies with a higher risk of bias and those with the highest weighting. The association between SGLT2 inhibitors (versus other non-SGLT2-inhibitor antihyperglycemic agents) and DKA remained significant (pooled HR = 1.51, 95% CI = 1.38-1.66, $P < 0.001$). A symmetrical funnel plot and an insignificant Egger's test value ($t = -0.68$, $P = 0.507$) indicated the absence of publication bias (refer to Supplementary Figure 1).

Table 3
Meta-regression analysis

Variables	Analyses 1: Using 21 cohorts (ACNU + prevalent NU)	Analyses 2: Using 20 cohorts (ACNU only)
	<i>P</i> -value	<i>P</i> -value
Study location	0.02	0.01
Analysis principle	<0.0001	<0.0001
Canagliflozin as dominant SGLT2i molecule	0.04	0.24
Exclusion of patients with some chronic illness	0.007	<0.0001

Abbreviation used: ACNU, active comparator new user design; Prevalent NU, prevalent new user design.

Heterogeneity evaluation

The cohorts observed were highly heterogeneous ($I^2 = 76\%$, $P < 0.01$). Removing the one study that employed the prevalent new user design⁶⁵ only slightly reduced the heterogeneity ($I^2 = 72\%$, $P < 0.001$).

Discussion

In this systematic review, a meta-analysis of nearly 3 million adult patients with T2DM from 14 observational studies across diverse geographical regions revealed a real-world positive association between SGLT2 inhibitors and DKA. Before the introduction of SGLT2 inhibitors, no other drug class was known to consistently lead to DKA.⁶⁹ Real-world data on the incidence of

SGLT2-inhibitor-associated DKA is crucial; and a better understanding of the underlying mechanisms and risk factors is essential to develop safety measures to prevent DKA. In recent years, dapagliflozin has been approved for the treatment of chronic kidney disease,^{53,70} and empagliflozin has received expanded approval for use in adults with heart failure.⁷¹ The New Zealand Society for the Study of Diabetes recommends SGLT2 inhibitors and GLP-1RA as ideal second-line medications in all patients with T2DM regardless of glycemic control or any other antihyperglycemic treatment. SGLT2 inhibitors are preferred over GLP-1RA in renal disease and heart failure comorbidities.⁷² As the use of SGLT2 inhibitors continues to expand and their importance grows, it is crucial to closely examine their real-world use and identify risk factors that may lead to severe complications.

Table 4
Leave-one-out sensitivity tests and conservative estimation of DKA risk

Model	Analyses 1: Using 21 cohorts (ACNU + prevalent NU)				Analyses 2: Using 20 cohorts (ACNU only)			
	Summary HR*		Heterogeneity*		Summary HR*		Heterogeneity*	
	Pooled HR (95% CI)	<i>P</i> -value	I^2	<i>P</i> -value	Pooled HR (95% CI)	<i>P</i> -value	I^2	<i>P</i> -value
Model 1: SGLT2i versus all comparators	1.33 (1.14, 1.55)	0.0003	76%	<0.001	1.28 (1.11, 1.47)	0.0006	72%	<0.001
Repeat model 1 analysis excluding 1 cohort with high weightage contribution – Dawwas et al ⁵⁶ (Cohort I: SGLT2i versus DPP4i comparator)	1.31 (1.12, 1.55)	0.001	76%	<0.001	1.26 (1.09, 1.45)	0.002	72%	<0.001
Repeat model 1 analysis excluding 1 cohort with high weightage contribution – Paterno et al ⁶² (Cohort II: without baseline CVD)	1.30 (1.11, 1.53)	0.001	75%	<0.001	1.25 (1.08, 1.44)	0.002	68%	<0.001
Repeat model 1 analysis excluding 1 cohort with high weightage contribution – Dawwas et al ⁵⁶ (Cohort II: SGLT2i versus SU comparator)	1.32 (1.12, 1.55)	0.001	77%	<0.001	1.26 (1.09, 1.46)	0.002	72%	<0.001
Repeat model 1 analysis excluding all three cohorts with the highest weightage contribution ^{56,62}	1.27 (1.06, 1.52)	0.008	73%	<0.001	1.19 (1.03, 1.39)	0.02	60%	0.001
Repeat model 1 analysis excluding all cohorts with high risk of bias ^{58,67,68}	1.60 (1.39, 1.85)	<0.0001	53%	0.01	1.55 (1.41, 1.71)	<0.0001	28%	0.17
Repeat model 1 analysis excluding all cohorts with high risk of bias ^{58,67,68} and all three cohorts with the highest weightage contribution ^{56,62}	1.58 (1.27, 1.98)	<0.0001	62%	0.004	1.51 (1.38, 1.66)	<0.0001	22%	0.23

Abbreviations used: ACNU, active comparator new user design; CVD, cardiovascular disease; DKA, diabetic ketoacidosis; DPP4i: dipeptidyl peptidase 4 inhibitors; GLP-1RA: glucagon-like peptide 1 receptor agonists; HR, hazard ratio; Prevalent NU, prevalent new user design; SGLT2i, sodium-glucose transporter-2 inhibitors; SU, sulfonylureas.

At the start of the previous decade, systematic reviews of RCTs did not establish a relationship between the SGLT2-inhibitor class of medications and DKA.^{68,73,74} However, more recent reviews that collectively analyzed 57 RCTs and 5 cohort studies have found that SGLT2 inhibitors significantly increased the risk of DKA, by nearly twofold.^{33–36} This is reflected in our review of real-world data with a pooled hazard ratio of 1.33. Our review confirms the suspected increased real-world DKA risk found in the first systematic review of population-based observational studies on SGLT2 inhibitors' safety in 2021.³⁷ However, while we report a significant relative risk of DKA, our results reveal a low baseline risk of DKA among new users of SGLT2 inhibitors, ranging from 0.6 to 6.3 per 1000 person-years. This underscores the importance of a comprehensive risk assessment to inform clinical decision-making on the part of healthcare providers.

This review found that studies conducted in North America and Scandinavia showed an increased risk of DKA with SGLT2-inhibiting treatment, while studies from Asia had a neutral risk association. These findings highlight the need for careful monitoring of patients who use SGLT2 inhibitors, especially in regions where the risk is higher. The racial/ethnic composition of the study populations included in our review was specified in only 5 studies.^{56–58,62,63} A recent large postmarketing observational study found the prevalence of DKA among Japanese patients treated with ipragliflozin to be low,⁷⁵ while in New Zealand a study reported a lower risk of DKA in non-European patients with T2DM taking empagliflozin,⁷⁶ which is consistent with our findings. Although racial disparities in DKA prevalence have been reported in past studies involving patients with T1DM and T2DM,^{77,78} they have not been fully explored in patients with T2DM treated with SGLT2 inhibitors.⁷⁹ Future research in this area can directly inform clinical practice by helping to adjust treatment strategies or provide better counselling for higher-risk patients.

Based on a comprehensive evaluation of the evidence, our analysis indicates that canagliflozin increases the risk of DKA. This conclusion contradicts the view that the risk applies to the entire drug-class as previously noted by other studies.^{56,65} Future studies should determine any potential variations in risk among the SGLT2-inhibiting drugs to gain a better understanding of the safety profiles of these novel medications. In this review, some studies employed criteria to exclude patients with specific chronic conditions such as cancer, kidney disease, HIV, organ transplant, and end-stage illness. From a clinical standpoint, excluding these illnesses is unlikely to result in a higher risk of SGLT2-associated DKA, which makes our finding to the contrary inconclusive.⁸⁰ Future research is required to examine the interplay between illness and the actual risk of DKA in patients with T2DM who take SGLT2 inhibitors.

Pharmacists have a responsibility to monitor the treatment strategies of patients with diabetes.⁸¹ By shedding light on the risk of DKA associated with SGLT2 inhibitors, this study equips pharmacists with the essential knowledge to enhance patient care, safety, and communication.

Limitations

The primary strength of our review is that the evidence is mainly derived from noncommercially funded, large- and medium-sized, ACNU-designed cohorts, offering new insights

that supplement evidence from RCTs and reflect real-world practice.^{42,82} However, our findings have several limitations. High heterogeneity was found among the studies included. This was moderated by the geographical origin of the observed population, the principle of analysis employed to define follow-up terms, and certain comorbidities. Studies that censored on drug-change/-discontinuation (AT and PP designs) reported a significant positive association between SGLT2 inhibitors and DKA, while the opposite was noted with ITT analysis. ITT is described as a “once randomized, always analyzed” analysis principle.⁸³ AT or PP statistical approaches can introduce potential confounding bias by the lack of randomization. This undermines the generalizability of our findings with regard to a positive SGLT2-inhibitor-DKA association because non-adherence to medications is very common among the patients with T2DM.⁸⁴ Although all of the studies included in this review utilized 1:1 propensity score matching and included an active comparator to eliminate indication bias, there were variations in the number of confounders considered or reported and the rigor of the statistical analysis. We believe that this meta-analysis successfully demonstrated that no single study influenced our conclusions. Additionally, none of the studies in this review diagnosed DKA based on biochemical data, but instead identified DKA using codes from the International Classification of Diabetes, ninth or 10th edition (ICD-9-CM and ICD-10-CM). This may have led to the misclassification of DKA cases.⁸⁵ While we conducted a systematic search for eligible studies with the help of an in-house librarian, it is possible that we missed studies that were not in English or not yet published. We also chose to exclude non-peer-reviewed publications to maintain confidence in the overall findings. Including studies with different designs or using a placebo instead of an active comparator may have increased the number of the cohorts examined, but it would have also led to greater heterogeneity and decreased confidence in the conclusions due to unresolved biases and remaining confounders.

Conclusion

This systematic review and meta-analysis of 21 cohorts from mostly noncommercial large and medium-sized, ACNU-designed studies provides an updated insight into the real-world prevalence of DKA with SGLT2 inhibitor treatment. Our findings agree with systematic reviews of RCTs and suggest a higher risk of DKA among T2DM adult patients taking SGLT2 inhibitors (versus other antihyperglycemic agents). Our findings serve to highlight the rarity but real hazard of SGLT2-inhibitor-associated DKA and call for continued vigilance from the part of caregivers, policymakers, and patients with diabetes through individualized monitoring of the ketone bodies in the blood.

Disclosure

The authors declare no conflict of interest.

Acknowledgments

The authors thank Berit Anderson for her guidance in conducting database searches.

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Appendix

Supplement to: Al-Hindi, B., Mohammed, M.A., Mangantig, E., & Martini, N.D. Prevalence of sodium-glucose transporter 2 inhibitor-associated diabetic ketoacidosis in real-world data: A systematic review and meta-analysis.

Supplementary Table 1
Search terms used in MEDLINE search

Process no.	Searches	Results
1	Exp Ketosis/	9586
2	(Acetonemia* or Acetonuria* or Ketoacid* or Ketonemia* or Ketonuria* or Ketosis or Acetone bodies or ketone bodies).mp. [mp=title, book title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms]	24,242
3	1 or 2	24,242
4	Exp sodium-glucose transporter 2 inhibitors/	4914
5	(Gliflozin* or sgl2 inhibitor* or sgl2-inhibitor* or sgl2 inhibitor* or sodium glucose transporter 2 inhibitor* or sodium-glucose transporter 2 inhibitor* or SLC5A2 or canagliflozin or Invokana or dapagliflozin or Farxiga or Forxiga or empagliflozin or Jadiance or Ertugliflozin or Steglatro or ipragliflozin or Suglat Luseogliflozin or Lusefi or Remogliflozin or Sergliflozin or Sotagliflozin or Zynquista or Tofogliflozin or Apleway or Deberza or Bexagliflozin or Henagliflozin or Licogliflozin).mp. [mp=title, book title, abstract, original title, name of substance word, subject heading word, floating sub-heading word, keyword heading word, organism supplementary concept word, protocol supplementary concept word, rare disease supplementary concept word, unique identifier, synonyms]	8795
6	4 or 5	8795
7	3 and 6	783

Supplementary Table 2

Summary of included studies

Exposure/Serial no.	Study summary
Empagliflozin 1.1	A large 2022 American study reported a positive association between empagliflozin (versus. DPP4i) and DKA, with wide CIs. ⁵⁷
SGLT2i Class 1.2	A large 2022 American study reported a positive association between SGLT2i (versus. both DPP4i and sulfonylureas) and DKA, with narrow CIs. ⁵⁶ molecule-specific risk estimates supported a class effect due to higher DKA risk for each of canagliflozin, dapagliflozin and empagliflozin.
1.3	A medium-sized 2021 Singaporean study showed a neutral association between SGLT2i (versus. DPP4i) and DKA, with narrow CIs. ⁵⁸ Interestingly, SGLT2i (versus. DPP4i) had a negative association with DKA in Malay patients, suggesting an ethnicity factor to DKA risk.
1.4	A medium-sized 2021 American study reported a positive association between SGLT2i (versus. metformin) and DKA in treatment-naïve adults, with wide CIs. ⁵⁹
1.5	A large 2021 Canadian study demonstrated a positive association between SGLT2i (versus. DPP4i) and DKA, with wide CIs. ⁶⁰ prior insulin use and a baseline HbA1c above 10% (86 mmol/mol) were found to be important risk factors, increasing the odds of developing DKA.
1.6	A small 2021 Italian study reported a neutral association between SGLT2i (versus. DPP4i) and DKA. ⁶¹
1.7	A large 2021 American study showed a positive association between SGLT2i (versus. GLP-1RA) and DKA, with narrow CIs. ⁶²
1.8	A medium-sized 2021 Italian study reported very few and similar frequency of DKA events in SGLT2i users versus. DPP4i users. ⁵⁵
1.9	A large 2021 American study suggested a positive association between SGLT2i (versus. GLP-1RA) and DKA, with narrow CIs. ⁶³
1.10	A medium-sized 2021 South Korean study reported a neutral association between SGLT2i (versus. DPP4i) and DKA, with narrow CIs. ⁶⁴
1.11	A large 2020 Canadian study noted a positive association between SGLT2i (versus. DPP4i) and DKA, with wide CIs. ⁶⁵ molecule-specific risk estimates supported a class effect due to higher DKA risk for each of canagliflozin, dapagliflozin and empagliflozin. Interestingly, the risk was augmented in patients who had a less severe form of diabetes.
1.12	A large 2019 American study that observed a positive association between SGLT2i (versus. sulfonylureas) and DKA, with narrow CIs. ⁶⁷ DKA risk was lower in SGLT2i users versus. insulin users, and was similar in SGLT2i users versus. thiazolidinediones users.
1.13	A large 2019 Scandinavian study reported a positive association between SGLT2i class (versus. DPP4i) and DKA, with wide CIs. ⁵⁰
1.14	A large 2019 South Korean study reported a neutral association between SGLT2i class (versus. DPP4i) and DKA, with narrow CIs. ⁶⁶ non-significant increases in DKA risk were noted in SGLT2i users who took diuretics and those who had microvascular complications.
1.15	A medium-sized 2018 Scandinavian study showed a positive association between SGLT2i class (versus. GLP-1RA) and DKA, with wide CIs. ⁵¹
1.16	A large 2017 American study concluded that there was a neutral association between SGLT2i (versus. other antihyperglycemic agents) and DKA, with wide CIs. ⁶⁸

Abbreviations used: CI: confidence interval; DKA: diabetic ketoacidosis; DPP4i: dipeptidyl peptidase 4 inhibitor; GLP-1RA: glucagon-like peptide 1 receptor agonist; SGLT2i: sodium-glucose transporter 2 inhibitor; HbA1c: glycated hemoglobin.

Note: The qualitative terms 'small', 'medium-sized', and 'large' refer to studies that included less than 5,000, between 5000–20,000, and more than 20,000 SGLT2i new users, respectively.³²

Supplementary Table 3

Risk of bias assessment using JBI critical appraisal checklist for cohort studies

	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11
	Were the two groups similar and recruited from the same population?	Were the exposures measured similarly to assign people to both exposed and unexposed groups?	Was the exposure measured in a valid and reliable way?	Were confounding factors identified?	Were strategies to deal with confounding factors stated?	Were the groups/ participants free of the outcome at the start of the study (or at the moment of exposure)?	Were the outcomes measured in a valid and reliable way?	Was the follow up time reported and sufficient to be long enough for outcomes to occur?	Was follow up complete, and if not, were the reasons to loss to follow up described and explored?	Were strategies to address incomplete follow up utilized?	Was appropriate statistical analysis used?
Patorno et al (2022) ⁵⁷	Y	Y	Y	Y	Y	Unclear	Y	Y	Y	Y	Y
Dawwas et al (2022) ⁵⁶	Y	Y	Y	Y	Y	Y	Y	Y	N	Y	Y
Goh et al (2022) ⁵⁸	Y	Y	Y	Y	Y	Unclear	Y	N	N	Y	Y
Fralick et al (2021) ⁵⁹	Y	Y	Y	Y	Y	Unclear	Y	Y	Y	Y	Y
Fralick et al (2021) ⁶⁰	Y	Y	Y	Y	Y	Unclear	Y	Y	N	Y	Y
Longato et al (2021) ⁶¹	Y	Y	Y	Y	Y	Unclear	Y	Y	N	Y	Y
Patorno et al (2021) ⁶²	Y	Y	Y	Y	Y	Unclear	Y	Y	N	Y	Y
Baviera et al (2021) ⁵⁵	Y	Y	Y	Y	Y	Unclear	Y	Y	N	Y	Unclear
Patorno et al (2021) ⁶³	Y	Y	Y	Y	Y	Unclear	Y	Y	Y	Y	Y
Han et al (2021) ⁶⁴	Y	Y	Y	Y	Y	Unclear	Y	Y	N	Y	Y
Douros et al (2020) ⁶⁵	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Wang et al (2019) ⁶⁷	Y	Y	Y	Unclear	Y	Y	Y	Unclear	N	Y	Unclear
Pasternak et al (2019) ⁵⁰	Y	Y	Y	Y	Y	Unclear	Y	Y	N	Y	Y
Kim et al (2018) ⁶⁶	Y	Y	Y	Y	Y	Y	Y	Y	N	Y	Y
Ueda et al (2018) ⁵¹	Y	Y	Y	Y	Y	Unclear	Y	Y	N	Y	Y
Wang et al (2017) ⁶⁸	Y	Y	Y	Unclear	Y	Y	Y	N	N	Y	Unclear

Real-world evidence of SGLT2i-associated DKA

Supplementary Table 4
RISMA 2020 checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	TitlePage
ABSTRACT			
Abstract	2	See the PRISMA 2020 for abstracts checklist.	Done, included in these supplementary materials
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	p.3
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	p.4
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	p.5 & p.8
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	p.4
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Table 1 of supplementary materials
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	p.5-6
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	p.5-6
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (eg for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	p.5
	10b	List and define all other variables for which data were sought (eg participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	p.4 & p.13
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	p.6
Effect measures	12	Specify for each outcome the effect measure(s) (eg risk ratio, mean difference) used in the synthesis or presentation of results.	p.8
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (eg tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	p.6 & p.8
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	p.5
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	p.5 & p.6
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	p.6
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (eg subgroup analysis, meta-regression).	p.6
Reporting bias assessment	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	p.6
	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	p.6
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	p.6
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Figure 1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	N/A
Study characteristics	17	Cite each included study and present its characteristics.	Table 2
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Supplementary Table 3
Results of individual studies	19	For all outcomes, present, for each study: (a) Summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (eg confidence/credible interval), ideally using structured tables or plots.	Figure 2
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	p.9 & p.10
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (eg confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	p.8-10, & Figure 2

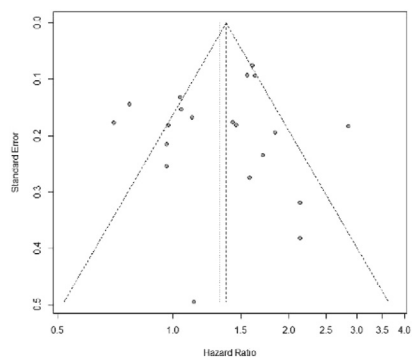
Supplementary Table 4 (continued)

Section and Topic	Item #	Checklist item	Location where item is reported
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	p.9 & Table 3
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	p.10 & Table 4
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Supplementary Table 3
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	p.8 & 10
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	p.11, paragraph 2
	23b	Discuss any limitations of the evidence included in the review.	p.12
	23c	Discuss any limitations of the review processes used.	p.12
	23d	Discuss implications of the results for practice, policy, and future research.	p.11-12 7 conclusion
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	p.4 & p.13
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	p.4
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	N/A
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	p.13
Competing interests	26	Declare any competing interests of review authors.	p.13
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: Template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	p.13

Supplementary Table 5

RISMA 2020 for abstracts checklist

Section and Topic	Item #	Checklist item	Reported (Yes/No)
TITLE			
Title	1	Identify the report as a systematic review.	Yes
BACKGROUND			
Objectives	2	Provide an explicit statement of the main objective(s) or question(s) the review addresses.	Yes
METHODS			
Eligibility criteria	3	Specify the inclusion and exclusion criteria for the review.	Yes
Information sources	4	Specify the information sources (e.g., databases, registers) used to identify studies and the date when each was last searched.	Yes
Risk of bias	5	Specify the methods used to assess risk of bias in the included studies.	Yes
Synthesis of results	6	Specify the methods used to present and synthesise results.	Yes
RESULTS			
Included studies	7	Give the total number of included studies and participants and summarise relevant characteristics of studies.	Yes
Synthesis of results	8	Present results for main outcomes, preferably indicating the number of included studies and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (ie which group is favoured).	Yes
DISCUSSION			
Limitations of evidence	9	Provide a brief summary of the limitations of the evidence included in the review (eg study risk of bias, inconsistency and imprecision).	Yes
Interpretation	10	Provide a general interpretation of the results and important implications.	Yes
OTHER			
Funding	11	Specify the primary source of funding for the review.	Not applicable
Registration	12	Provide the register name and registration number.	Yes



Supplementary Figure 1. Funnel Plot.